CHAPTER-1

INTRODUCTION

In today's fast-paced and competitive hospitality industry, effective hotel management is essential to ensure high-quality service and customer satisfaction. Traditional methods of managing hotel operations—such as manual record-keeping and paper-based bookings—are often time-consuming, error-prone, and inefficient. To address these challenges, the Hotel Management System has been developed as a digital solution to automate and optimize hotel operations.

The primary goal of this project is to create an integrated system that simplifies key hotel processes, including room reservations, guest check-in/check-out, billing, customer data management, and housekeeping coordination. By digitizing these tasks, the system enhances accuracy, speeds up service delivery, and provides hotel staff with real-time access to critical information.

This project not only reduces the workload of hotel employees but also improves the guest experience by ensuring smooth and efficient service. Whether implemented in a small lodge or a large resort, the Hotel Management System offers scalable functionality and a user-friendly interface, making it a valuable tool in modern hotel administration.

CHAPTER 2

LITERATURE SURVEY

Sr No	Paper Title	Author	Published Year	Methodology	Finding of the Paper
1.	A Study on Hotel Management System Using Cloud Computing	N. Rahul, V. Harsha, G. Mahesh	2020	Used web-based PHP and MySQL system to authenticate students and generate downloadable bonafide certificates.	Successfully reduced manual workload and improved the speed and accuracy of certificate issuance in colleges.
2.	The Impact of Hotel Management Information Systems Customer Satisfaction	R. Ramesh, S. Dinesh	2021	Developed an online platform using HTML, CSS, JavaScript, PHP, and MySQL. Implemented admin and student modules.	The system allowed real-time data verification and instant certificate download, eliminating paperwork and delays.
3.	Smart Hotel Management System Using IoT	Priya Sharma, Deepak Bansal	2019	Proposed a secure web application using Django and SQLite for managing academic certificate records.	Demonstrated efficient and tamper-proof generation and storage of academic certificates using role-based access control.
4.	Hotel Management System Based on Artificial Intelligence	Meenal K., Pavan T.	2022	Used PHP and cloud-based document verification through OCR for extracting data from student records.	Improved accuracy in certificate verification using OCR, beneficial for automating certificate issuing processes in colleges.

CHAPTER 3

DESIGN

3.1 Motivation

In most educational institutions, obtaining a hotel management manually is a time-consuming and tedious manual process. Students often need to fill out physical forms, wait in long queues, and follow up multiple times with the college office, which delays their academic or personal needs. Administrative staff, on the other hand, are overburdened with repetitive certificate requests that reduce their productivity and cause errors.

This inefficiency inspired us to develop a Hotel Management System that aims to digitize and automate the entire process.

- Simplify and speed up the bill generation process for customers.
- Reduce manual workload on administrative staff and improve their efficiency.
- Ensure data accuracy by using uploaded verified customer records.
- Improve accessibility, allowing customers to request and receive bills from anywhere.
- Promote paperless processes, contributing to sustainability and modernization of hotel institutions.

3.2 Problem Statement

In many hotels, especially small to mid-sized establishments, the reliance on manual processes for managing daily operations—such as room reservations, guest check-ins/check-outs, billing, and staff coordination—leads to inefficiencies, errors, and poor customer service. These outdated methods often result in overbooking, data loss, and difficulty in tracking guest history and financial records.

Moreover, existing software solutions in the market are often too expensive, overly complex, or not customizable to suit the specific needs of individual hotels. There is a pressing need for an affordable, user-friendly, and scalable hotel management system that can automate these operations, reduce human error, and enhance both administrative control and customer satisfaction.

3.3 Objective

- To automate Hotel Operations .
- To Improve Customer Service .
- To Efficient Data Management .
- To Enhance Operational Efficiency .
- To Provide Real-Time Information .
- To User-Friendly Interface .
- To Generate Reports and Analytics .
- To Role-Based Access Control .

3.4 Scope

1. **Room Booking and Reservation Management:**

   Users can book, cancel, and modify room reservations with real-time availability status.

2. **Check-In and Check-Out Processing:**

   Efficient handling of guest arrivals and departures, including automatic room status updates.

3. **Customer Information Management:**

   Secure storage and retrieval of guest data, including personal information, preferences, and booking history.

4. **Billing and Payment Processing:**

   Generation of invoices, handling of multiple payment methods, and automatic tax calculations.

5. **Room and Inventory Management:**

   Tracking of room status (available, occupied, under maintenance) and management of basic hotel inventory.

6. **Staff Role and Access Control:**

   Role-based access for admin, receptionist, and housekeeping to ensure security and data integrity.

7. **Report Generation:**

   Automated generation of reports for reservations, revenue, occupancy rates, and customer statistics.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 Hardware Requirements

- Processor: Intel i3 or higher (or AMD equivalent)
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: At least 500 MB of free space
- Display: Minimum resolution 1366 x 768
- Network: Localhost server setup (LAN) or Internet access for deployment

4.2 Software Requirements

- Operating System: Windows 10 or later / Linux / macOS
- Web Server: XAMPP (Apache, MySQL, PHP)
- Backend: PHP 7.4 or higher
- Frontend: HTML5, CSS3, JavaScript, Bootstrap
- Database: MySQL (via phpMyAdmin)
- Libraries/Packages:
- PhpOffice\PhpWord (for reading DOCX files)
- Smalot\PdfParser (for parsing PDF files)
- Composer (Dependency manager for PHP)
- Browser: Google Chrome / Mozilla Firefox / Microsoft Edge
- 8. Code Editor: VS Code / Sublime Text / Notepad++

CHAPTER 5

SYSTEM ARCHITECTURE

5.1 SYSTEM ARCHITECTURE

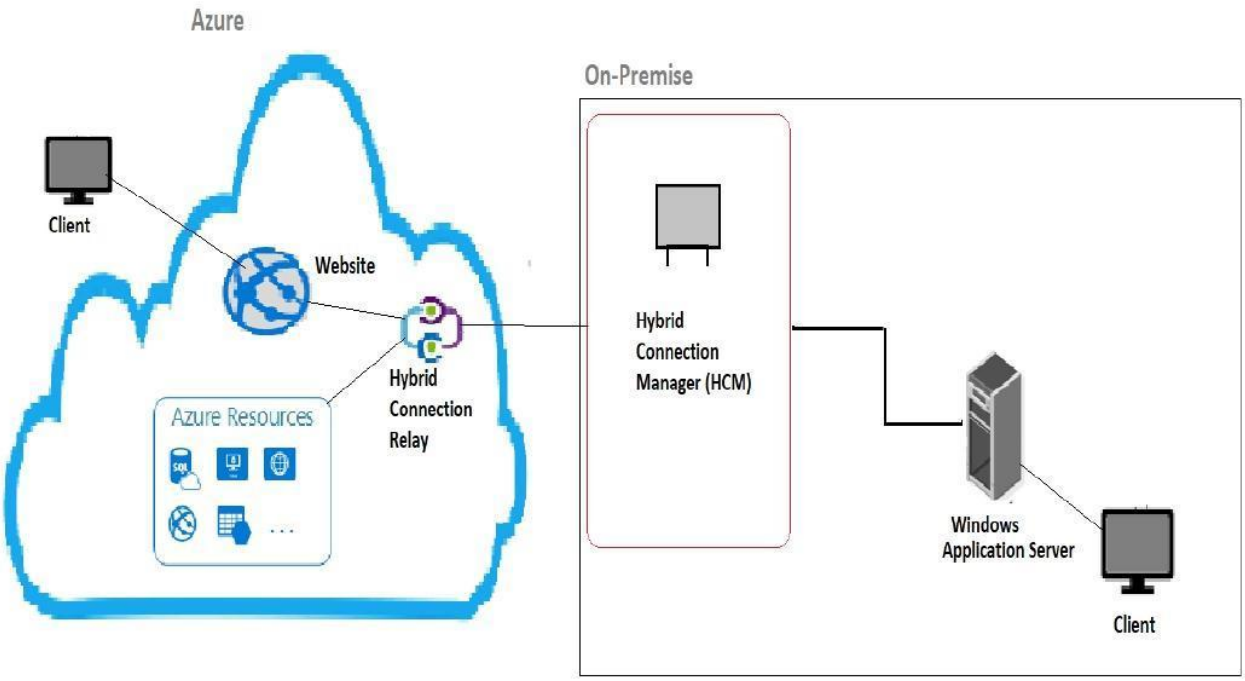
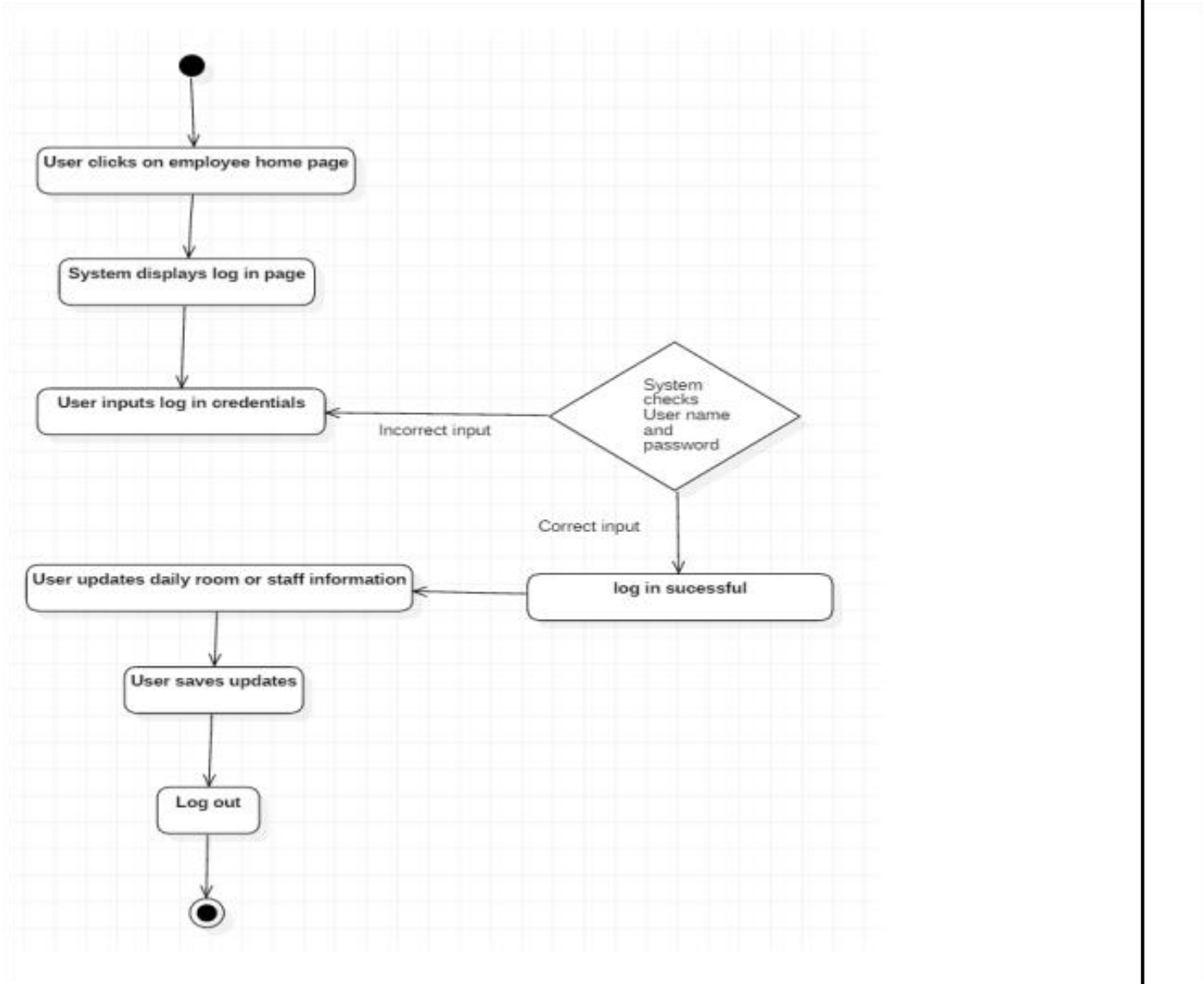


FIGURE 1. Hybrid Server-Client Architecture of the Hotel management system

5.2 FLOWCHART



5.3 ALGORITHMS USED

Hotel Management System Algorithm

1. Initialize System

- Load hotel data (rooms, rates, bookings, etc.)
- Initialize user interface (UI)

2. Manage Rooms

- Display current room status (available, booked, maintenance)
- Add Room
- Input room number, type, rate, status
- Save room details

3. Update Room

- Display current reservations
- Create Reservation
- Check room availability
- Update details (dates, room type, guest info)

4. Check-In Guests

- Input reservation ID
- Verify reservation details
- Update room status to "occupied"

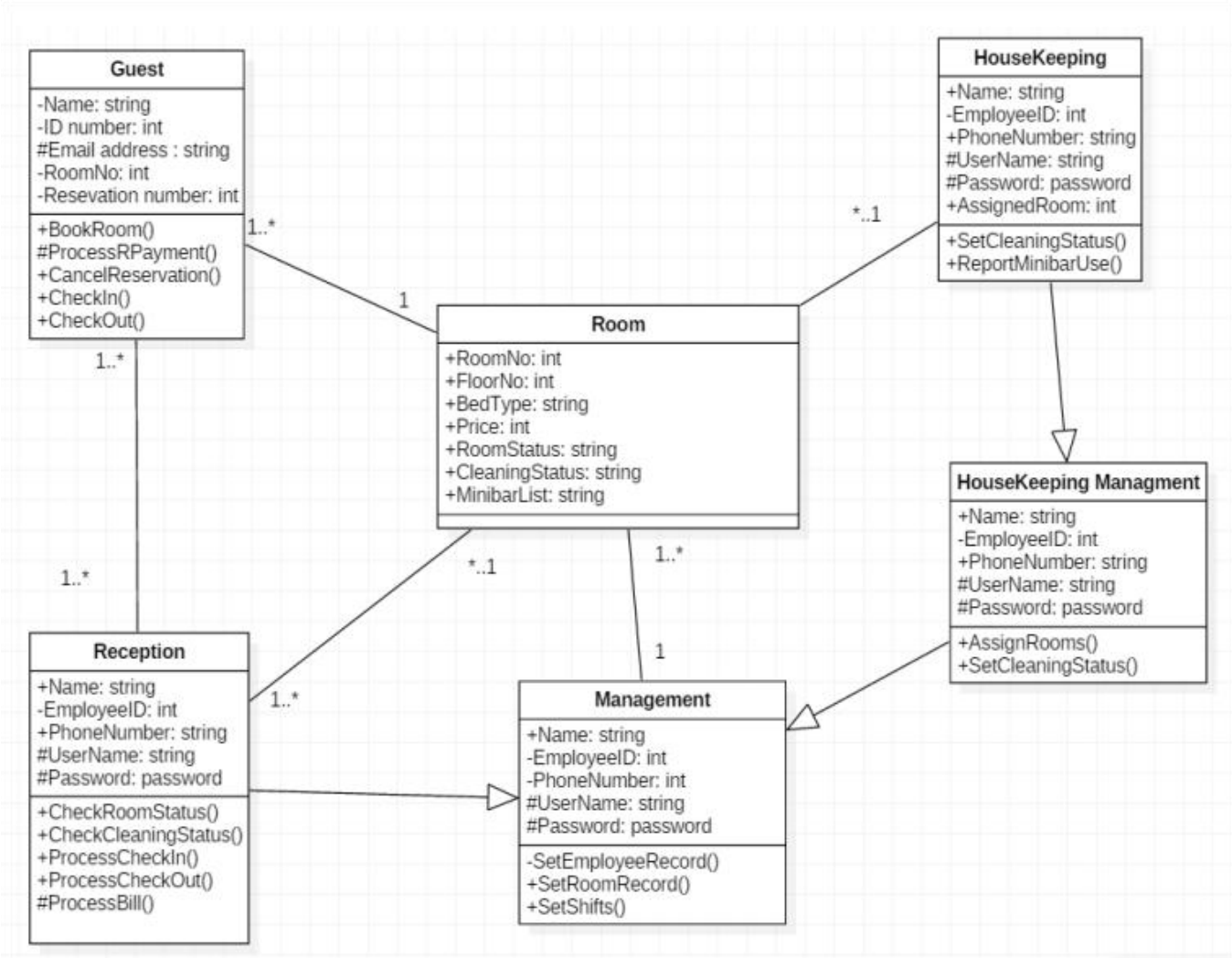
5. Generate Reports

- Daily Revenue Report
- Occupancy Report
- Reservation Report

6. Exit System

- Save all changes to the database
- Close the application

5.4 DATASET


Guest
-Name: string
-ID number: int
#Email address : string
-RoomNo: int
-Resevation number: int
+BookRoom()
#ProcessRPayment()
+CancelReservation()
+CheckIn()
+CheckOut()
HouseKeeping
+Name: string
-EmployeeID: int
+PhoneNumber: string
#UserName: string
#Password: password
+AssignedRoom: int
+SetCleaningStatus()
+ReportMinibarUse()
Room
+RoomNo: int
+FloorNo: int
+BedType: string
+Price: int
+RoomStatus: string
+CleaningStatus: string
+MinibarList: string
HouseKeeping Managment
+Name: string
-EmployeeID: int
+PhoneNumber: string
#UserName: string
#Password: password
+AssignRooms()
+SetCleaningStatus()
Reception
+Name: string
-EmployeeID: int
+PhoneNumber: string
#UserName: string
#Password: password
+CheckRoomStatus()
+CheckCleaningStatus()
+ProcessCheckIn()
+ProcessCheckOut()
#ProcessBill()
Management
+Name: string
-EmployeeID: int
-PhoneNumber: int
#UserName: string
#Password: password
-SetEmployeeRecord()
+SetRoomRecord()
+SetShifts()
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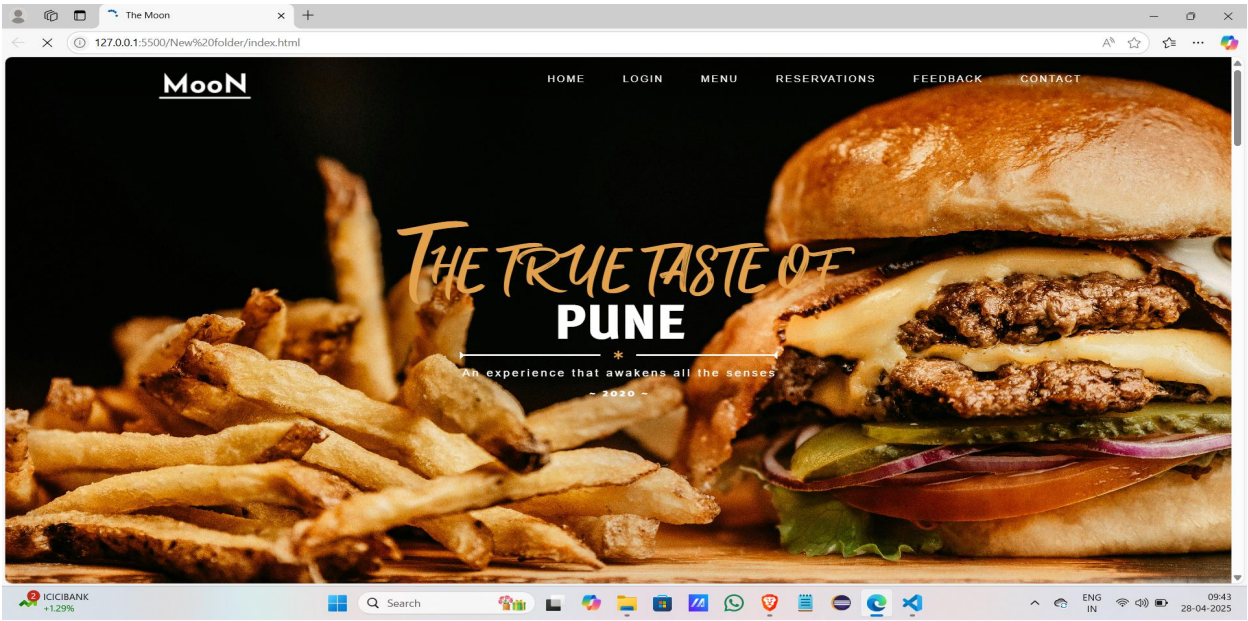
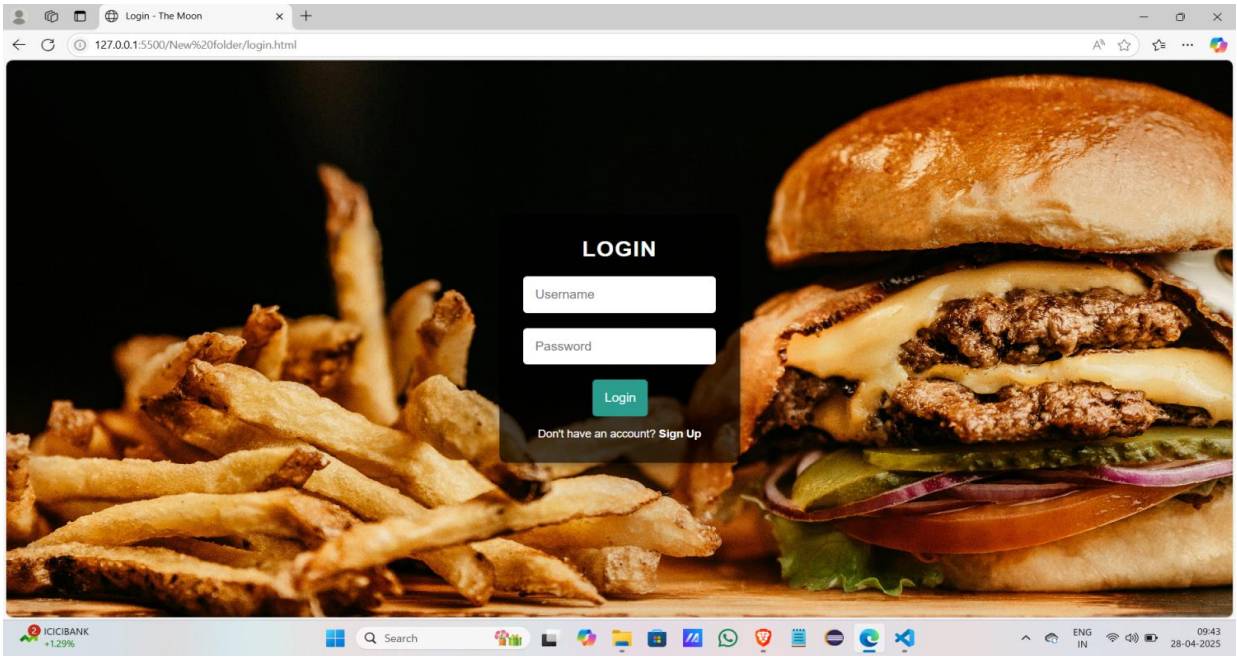
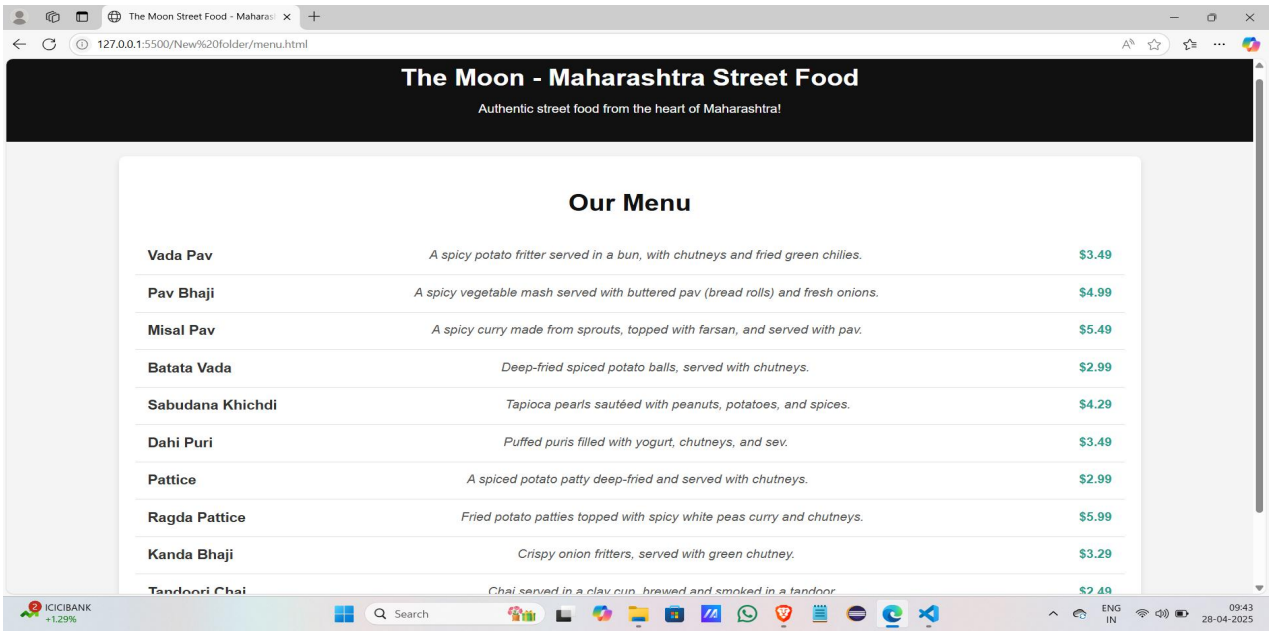
CHAPTER 6

RESULT AND ANALYSIS






The Moon Street Food - Maharas
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The Moon - Maharashtra Street Food
Authentic street food from the heart of Maharashtra!
Our Menu
Vada Pav | A spicy potato fritter served in a bun, with chutneys and fried green chilies. | $3.49
Pav Bhaji | A spicy vegetable mash served with buttered pav (bread rolls) and fresh onions. | $4.99
Misal Pav | A spicy curry made from sprouts, topped with farsan, and served with pav. | $5.49
Batata Vada | Deep-fried spiced potato balls, served with chutneys. | $2.99
Sabudana Khichdi | Tapioca pearls sautéed with peanuts, potatoes, and spices. | $4.29
Dahi Puri | Puffed puris filled with yogurt, chutneys, and sev. | $3.49
Pattice | A spiced potato patty deep-fried and served with chutneys. | $2.99
Ragda Pattice | Fried potato patties topped with spicy white peas curry and chutneys. | $5.99
Kanda Bhaji | Crispy onion fritters, served with green chutney. | $3.29
Tandoori Chai | Chai served in a clay cup, brewed and smoked in a tandoor | $2.49
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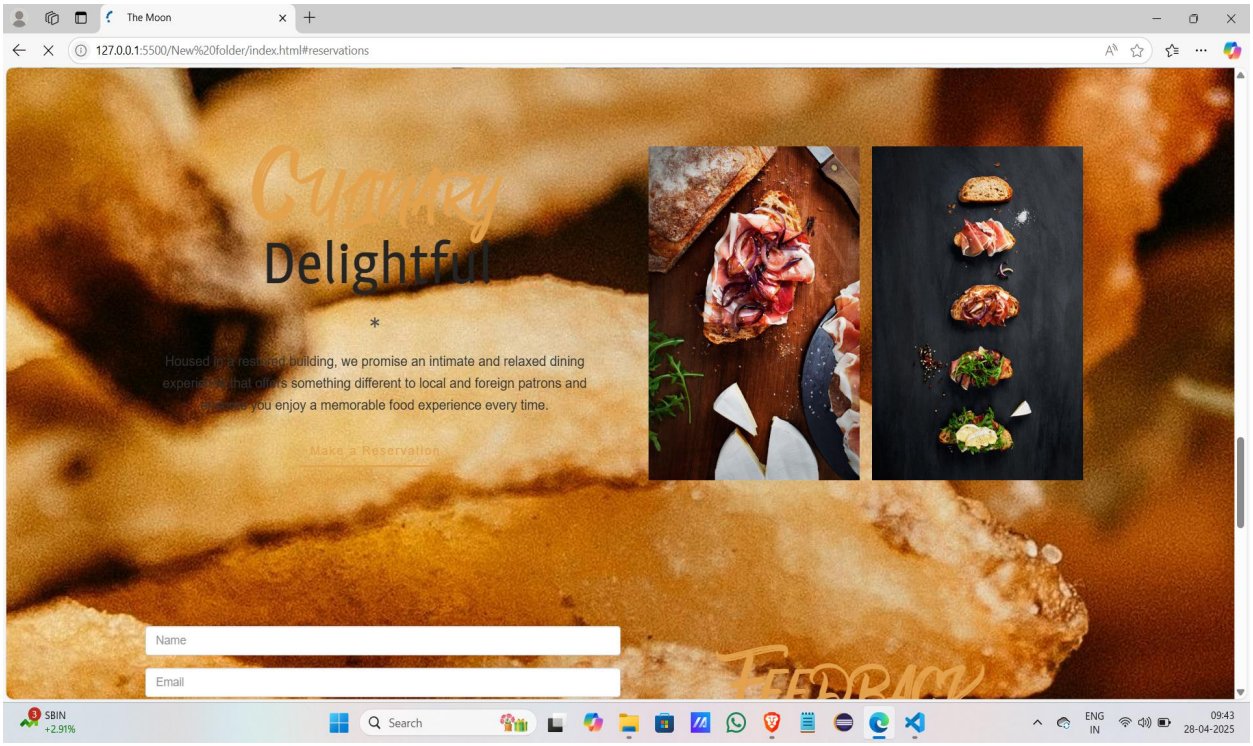
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CHAPTER 7

ADVANTAGES & DISADVANTAGES

7.1 Advantages:

- Automation of Routine Tasks
- Improved Accuracy
- Real-Time Room Management
- Centralized Data Storage
- Efficient Billing and Payment Processing
- Customizable Access Control

7.2 Disadvantages:

- Initial Setup Cost
- Training Requirements
- Dependence on Technology
- Data Security Risks
- System Downtime
- Maintenance and Updates
- Limited Customization in Some Solutions
- Risk of Data Loss

CHAPTER 8

APPLICATIONS AND FUTURE SCOPE

Application:

.• **Online Room Booking Portal**

Allows guests to view room availability, make reservations, and receive instant booking confirmations directly through the hotel's website.

- **Guest Profile Management**

Stores customer details and booking history, enabling personalized services and loyalty programs for returning guests.

- **Admin Dashboard for Hotel Staff**

Enables hotel managers and staff to oversee operations, view room statuses, manage bookings, and monitor revenue in real time from any device.

- **Billing and Invoicing System**

Automatically generates bills, handles multiple payment options (credit/debit cards, digital wallets), and stores transaction records.

- **Check-In/Check-Out Management**

Streamlines front desk processes by enabling digital check-in and check-out features, reducing wait times and paperwork.

Future Scope:

- **Integration with Online Travel Agencies (OTAs):**

Future versions can include APIs to sync bookings with platforms like Booking.com, Airbnb, and Expedia, enabling automatic room availability updates and reservation management.

· **Mobile App Support:**

A mobile-friendly version or dedicated app could allow both guests and staff to access the system on the go—for example, guests can check in remotely or staff can manage bookings from mobile devices.

· **Artificial Intelligence (AI) and Chatbots:**

AI-powered chatbots can assist with instant customer support, personalized recommendations, and automated booking responses 24/7.

· **Smart Room Integration (IoT):**

The system can be expanded to integrate with smart devices (e.g., keyless entry, climate control, smart lighting), offering a high-tech experience for guests.

· **Dynamic Pricing and Demand Forecasting:**

By using historical data and AI, the system could implement dynamic pricing models to maximize revenue based on demand, season, or events.

· **Multi-Language and Currency Support:**

To cater to international guests, future versions can offer multi-language interfaces and support for different currencies in billing.

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CHAPTER 9

CONCLUSION

The Hotel Management System project successfully demonstrates how technology can streamline and enhance the efficiency of hotel operations. By automating key processes such as room booking, guest check-in/check-out, billing, and reporting, the system reduces manual workload, minimizes errors, and improves service quality. It also provides a centralized platform for managing customer data, room status, staff roles, and financial records, contributing to better organization and decision-making.

The system is designed to be user-friendly, secure, and scalable, making it suitable for a wide range of hotels—from small lodges to larger establishments. While the current implementation covers essential hotel functions, there is significant potential for future enhancements such as mobile app integration, AI-driven analytics, and smart room technologies. Overall, this project lays a strong foundation for modernizing hotel management and delivering a better experience for both staff and guests.

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